APPLICATION OF LSTMS TO BITCOIN PRICE PREDICTION
LSTMS FOR (SHORT-HORIZON) BITCOIN PRICE FORECASTING (2015-2025)

Yilin Wang

Introduction

The cryptocurrency market is an emerging and rapidly evolving financial system characterized by a decentralized structure, continuous global trading, and relatively weak regulatory oversight.Unlike traditional assets such as stocks and bonds, cryptocurrency prices are more strongly influenced by global investor sentiment and technological innovation, leading to higher volatility and lower predictability (Gadirli et al., 2025). As the largest and most influential cryptocurrency, Bitcoin serves as the primary focus of this study.

The objective of this study is to deepen understanding of AI-driven forecasting in highly volatile financial markets and offer data-driven insights for investors and market analysts. This study uses the long short-term memory (LSTM)-based model to analyze and forecast Bitcoin's historical price data spanning a decade from 2015 to 2025 (Bitcoin Price Live, 2025). [2] The results indicate that the LSTM model is a viable and effective approach for predicting Bitcoin price trends.

Methodology

LSTM, as a deep learning model, demonstrates strong capability in capturing both short-term and long-term dependencies in time series data (Van Houdt et al., 2020). Each LSTM unit has a memory cell and gating mechanisms: input, forget and output gates. The structure of LSTM is shown in Figure 1.

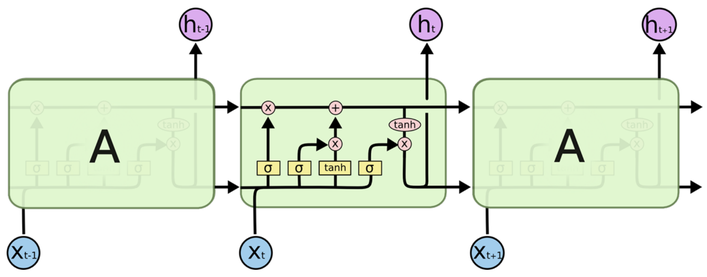


Figure 1. The Structure of LSTM

Forget Gate:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

determines which information from previous cell state C_{t-1} is discarded.

Input Gate and Candidate Cell State:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$
$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$$

controls which new information from xxx to be input to cell state. $\tilde{C}_t$ is potential new information to be added to the cell state.

Cell State Update

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$

combines old cell state with new candidate information, filtered by forget and input gates.

Output Gate and Hidden State

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$
$$h_t = o_t * \tanh(C_t)$$

decides how much of cell state information should influence output.

W_f is the weight matrix associated with f_t.
b_f is the bias term of the forget gate.
h_{t-1} represents the hidden state from the previous time step.
x_t is the current input vector.
$\sigma(\cdot) \in [0, 1]$ denotes the sigmoid activation function.

Data Analysis

Using daily Bitcoin prices (2015–2025), our analysis shows volatility clustering around major market cycles. Figure 2 indicates elevated swings in 2020–2022, followed by a strong post-2023 rebound

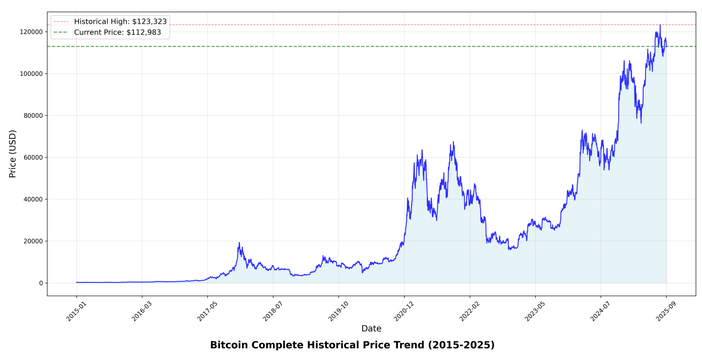


Figure 2. Historical Prices of Bitcoin (2015-2025)

Figure 3 shows a right-skewed distribution (mean $25,689 vs. median $10,785), while daily returns are nearly normally distributed within ±5%. Volatility patterns reveal significant temporal variation, with notable spikes during 2020-2022 and 2024-2025 market turbulence periods.

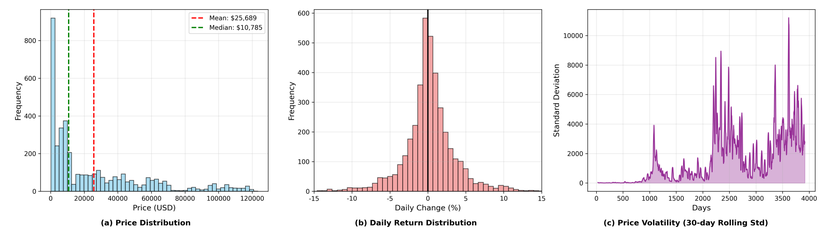


Figure 3. Summary statistics of Bitcon prices distribution, daily returns, and volatility (2015-2025)

train- test: 85% train and 15% test
Hyperparameter search: grid search: lookback window $L \in \{30, 60\}$, predict horizon $h \in \{3, 7\}$, learning rates $\xi \in \{0.0001/0.001\}$, LSTM units $U \in \{50/100\}$.
We split the data into 85% training and 15% test sets.
The scatter plot demonstrates a strong inverse relationship between R^2 and RMSE, with the optimal model (starred) achieving R^2=0.9656 and RMSE= 3,751



Figure 4. xxx

Figure 5 shows with predictions (orange) closely actual prices (blue) across major price movements peaks bottoms □□ LSTM's effectiveness for cryptocurrency price prediction.

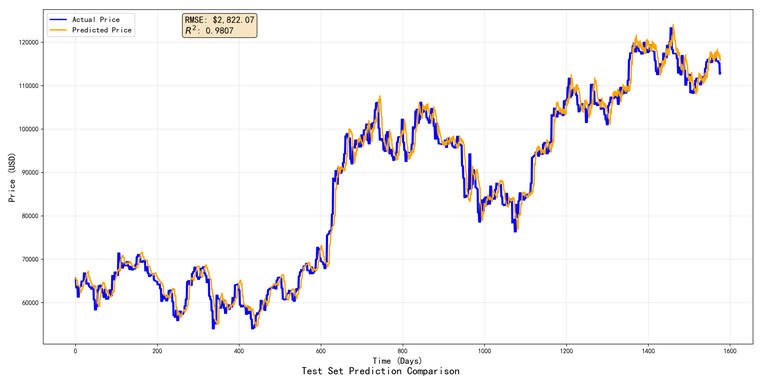
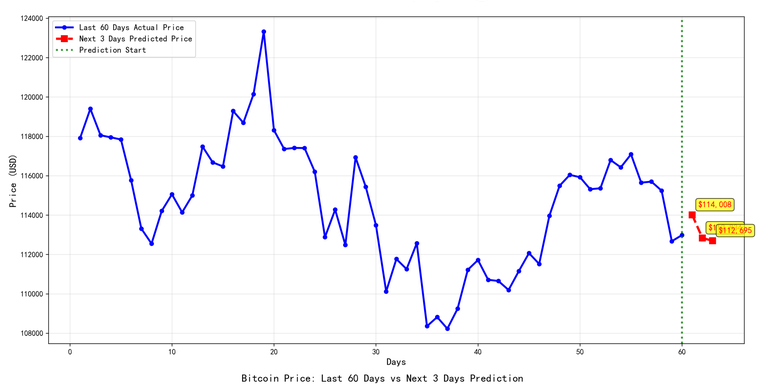


Figure 5. Predicted Prices generated by LSTM vs True Prices

The best LSTM model uses a 60-day lookback (blue) to predict the next three days (red): $114,008, $112,834, and $112,695, implying an overall 1.2% decline



Conclusion

This study examines whether LSTM can forecast short-term Bitcoin prices. We used an LSTM model with grid-searched hyperparameters on Bitcoin data (2015–2025). The optimized model (, showing LSTM is effective for short-term crypto forecasting. Future work should include volume and sentiment features for stronger robustness.

References

[1] Bitcoin (btc) price live - investing.com australia, 2024.

[2] Farid Gadirli, Min Chen, and Getachew A. Haile.
Investor attention and cryptocurrency volatility: A machine learning and econometric analysis.
`https://doi.org/10.2139/ssrn.5230632`, 2025.
SSRN Working Paper.

[3] G. Van Houdt, C. Mosquera, and G. Nápoles.
A review on the long short-term memory model.
Artificial Intelligence Review, 53(8), 2020.
Keep capitalization: Long Short-Term Memory (LSTM).